Physics 4730, Fall 2026 Homework #5

Due Sunday November 29th

1. Write your own implementation (*i.e.* don't use **scikit**) of the k Nearest Neighbors[1] (kNN) classification algorithm, as a Python function. The code should run on training and test sets in the format of the data supplied in **redblue.dat**, and provide predictions based on $k = 3$ nearest neighbors. Use the **scikit-learn** default 75/25 training/test split. Given the features of a particular "test" instance, the function should return the results of a binary classification decision *e.g.* "red" or "blue".

   (a) Extract a test dataset from **redblue.dat**. Apply your kNN algorithm to this test set and create a confusion matrix summarizing your results.

   (b) Now, modify your kNN function by increasing k from 3 to 5 and weighing terms according to the inverse square of the feature-space distance. Repeat the steps in Part (a) above.

   (c) In a separate markdown cell, write a quantitative comparison of the two kNN methods. In what ways are they similar and different?

   Submit your Jupyter Notebook `kNN.ipynb`.

2. Apply the **scikit-learn** kNN classification algorithm, with default parameters, to the full *Habitable Worlds Catalog* of exoplanets, with preprocessing data cuts as applied by the instructor in Lab 24.

   As in Lab 22, experiment with this method using both the two (stellar mass, orbital period) and three (mass, period, distance) feature classifications. In both cases, create a confusion matrix summarizing your results.

   Finally, use a separate markdown cell to comment quantitatively on the two- and three-feature analyses. In what ways are they similar and different?

   Submit your Jupyter Notebook `kNN_hwc.ipynb`.

Submit your assignment by 11:59 PM on Sunday, November 29:

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submit p4730 hw05 kNN.ipynb kNN_hwc.ipynb
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[1] T.M. Cover and P.E. Hart, IEEE Trans. Inf. Theory **13** 21 (1967).